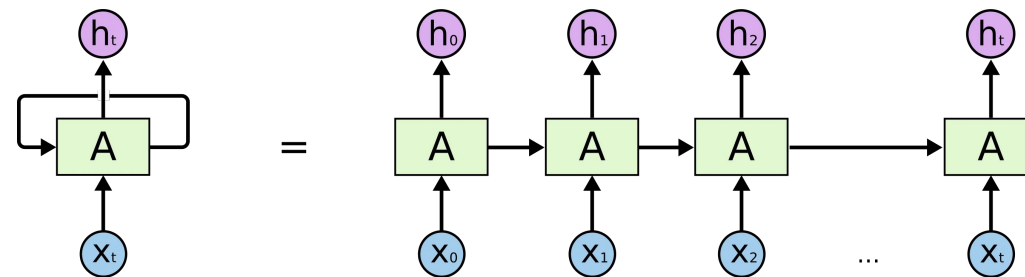


Recurrent Neural Networks

COMP6252: Deep Learning Technologies

University of Southampton UK



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VLC / ECS / UoS

Content

- Motivation
- RNN architectures
- GRU and LSTM

Examples of Sequence Data

- Speech recognition



“Deep learning is hard
but useful!”

- Music generation

ϕ



- Sentiment classification

“I like this movie!”



- DNA sequence analysis

ATTTGCGCAGAGACCTAAG



ATTTGCGCAGAGACCTAAG

- Machine translation

I like deep learning but do not
like mathematics



我喜欢深度学习但是
不喜欢数学

Motivating Example

➤ Name detection in text

Recall: words in $\mathbf{x}$ can be represented by one-hot encoding

Training samples



$\mathbf{x}$: Xiaohao and Hikmat have made the final exam closed-book

$\mathbf{y}$: 1 0 1 0 0 0 0 0 0

Length of $\mathbf{x}$: $T_x = 9$

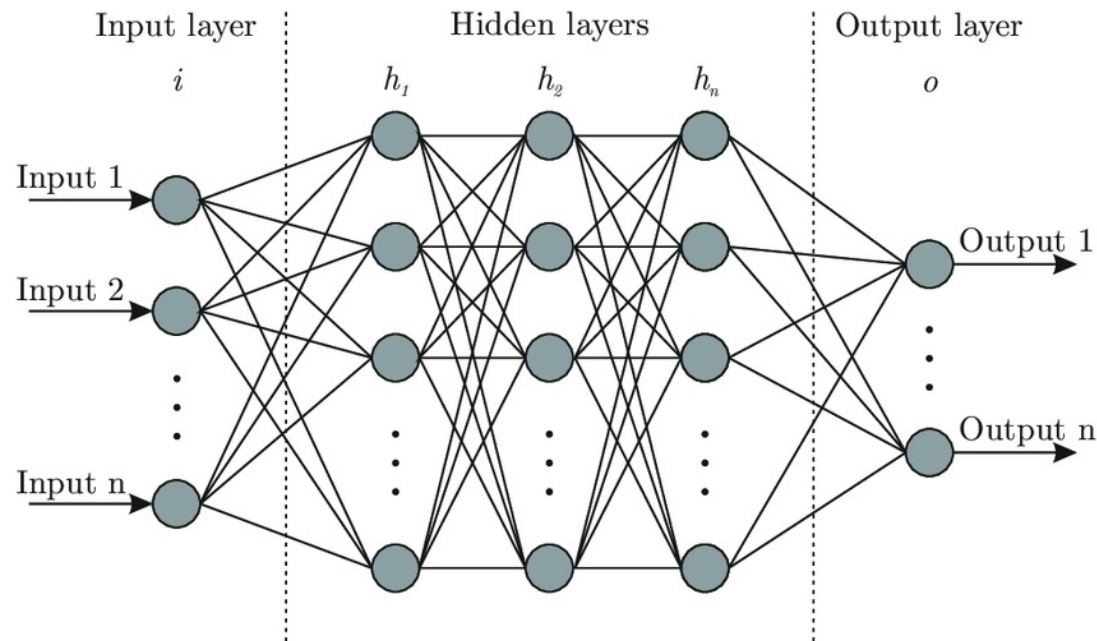
Length of $\mathbf{y}$: $T_y = 9$

T_x and T_y can be different across all the training samples

Motivating Example

➤ Name detection in text

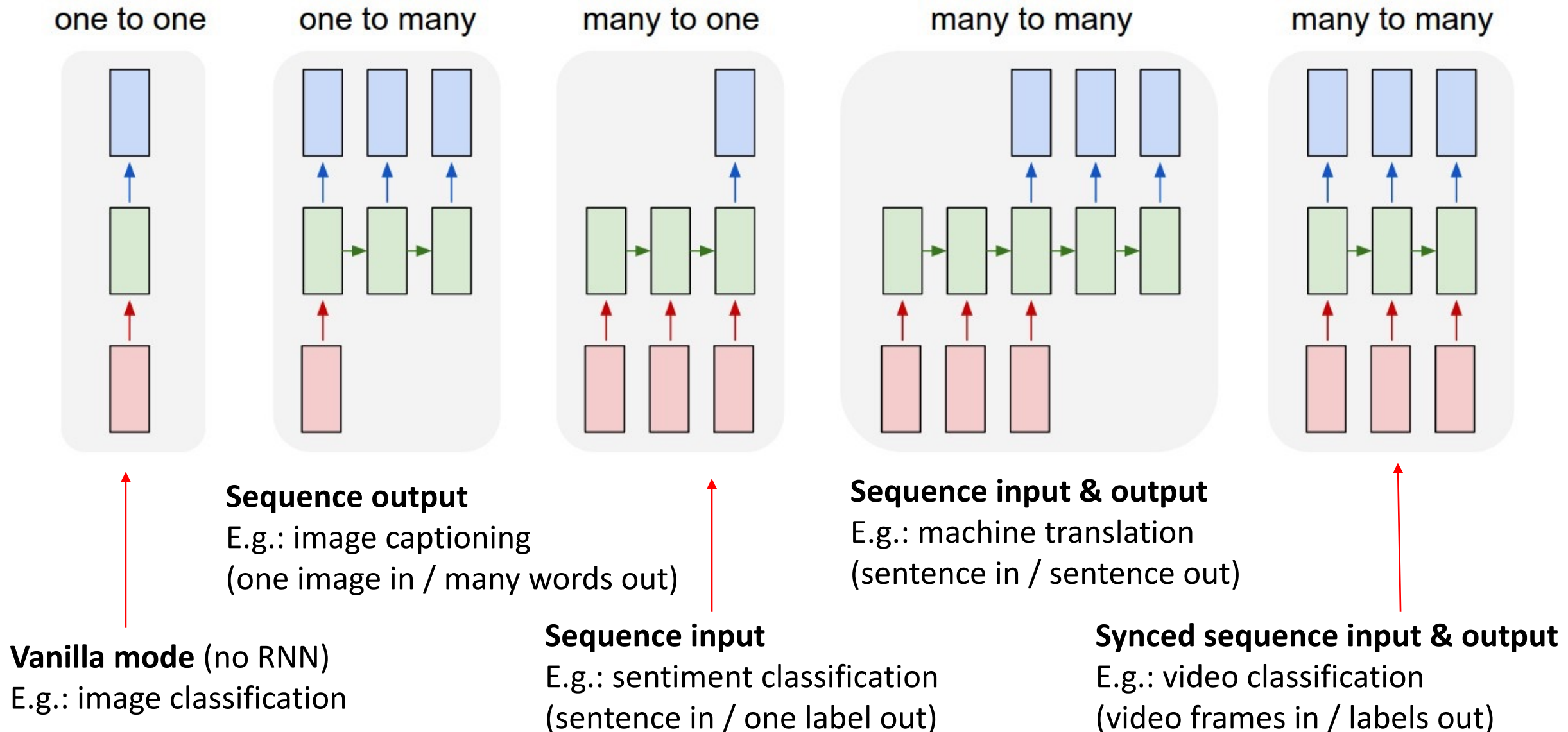
How about we use a **standard network** like:



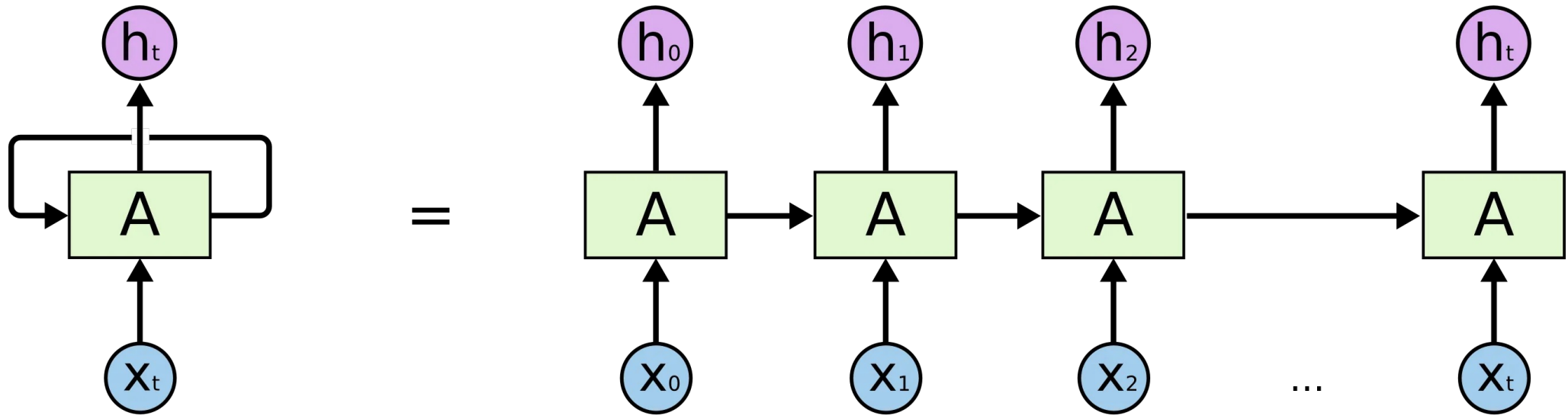
- Input and output can be **different** lengths in different examples
- **Doesn't share** features learned across different positions of text

- It is necessary to remember what happened **in the past**
- To facilitate this, we would like to add **a feedback loop** delayed in time

Recurrent Neural Network - Architectures



Recurrent Neural Network

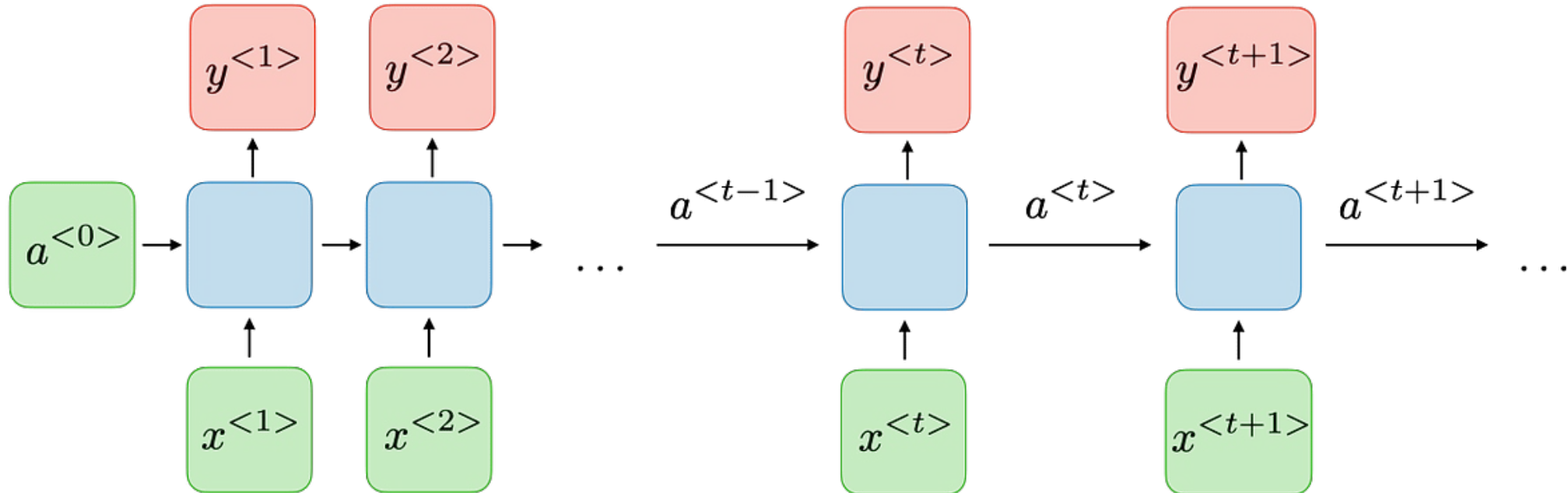


- RNNs are a family of ANNs for processing sequential data
- RNNs have directed cycles in their computational graphs

Recurrent Neural Network

x : Xiaohao and Hikmat have made the final exam closed-book

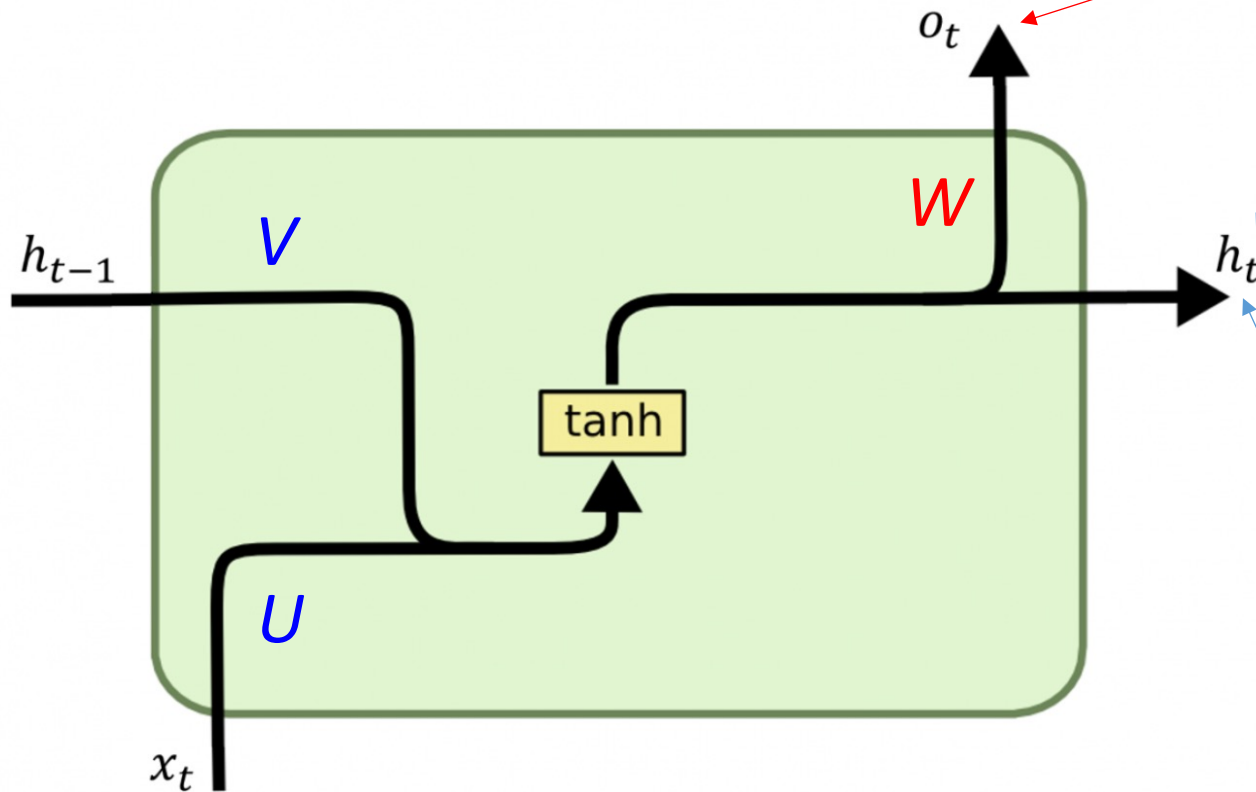
y : 1 0 1 0 0 0 0 0 0



Loss function of RNN, see e.g.:

<https://prvnk10.medium.com/loss-function-recurrent-neural-networks-rnns-ba4eb786d810>

Recurrent Neural Network



x_t : input vector ($m \times 1$).

h_t : hidden layer vector ($n \times 1$).

o_t : output vector ($n \times 1$).

b_h : bias vector ($n \times 1$).

U, W : parameter matrices ($n \times m$).

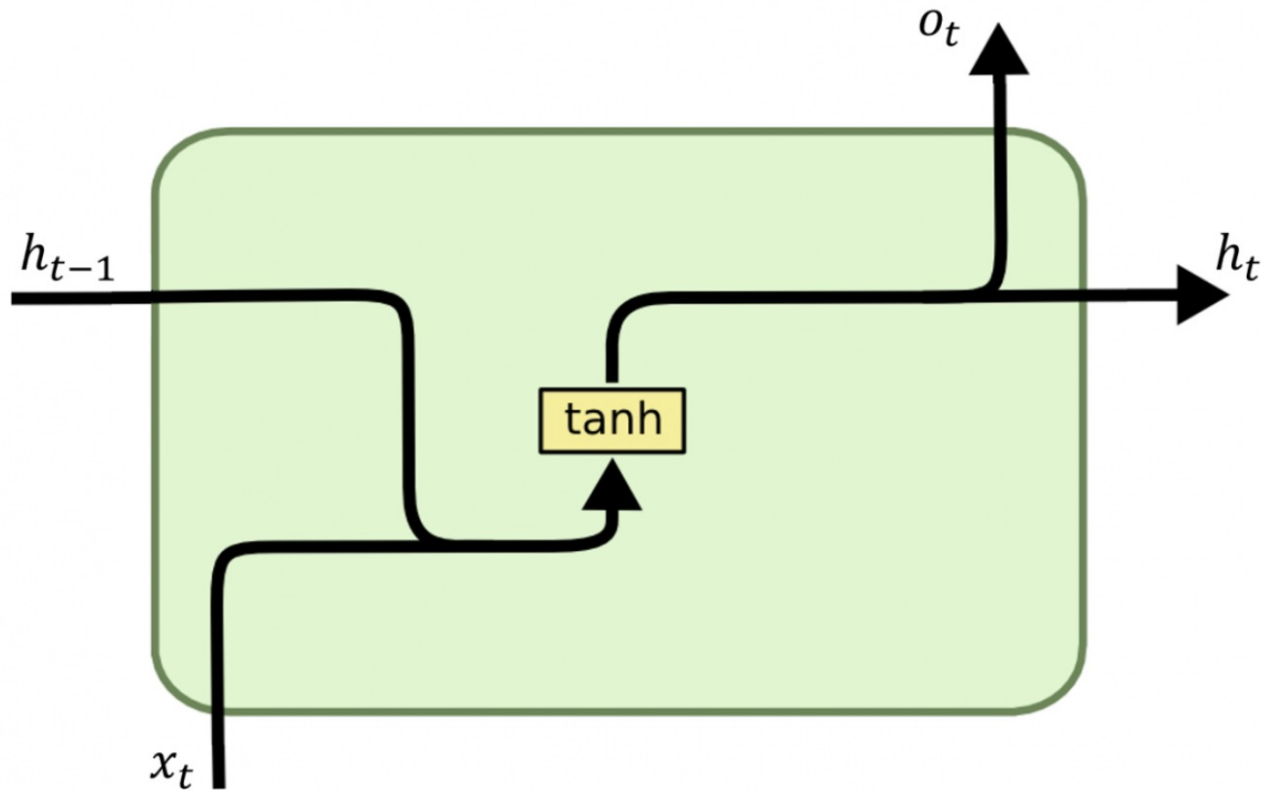
V : parameter matrix ($n \times n$).

σ_h, σ_y : activation functions.

$$h_t = \sigma_h(i_t) = \sigma_h(U_h x_t + V_h h_{t-1} + b_h)$$

$$y_t = \sigma_y(a_t) = \sigma_y(W_y h_t + b_h)$$

Recurrent Neural Network



Limitations:

Vanishing gradients

When a standard RNN network is exposed **to long sequences or phrases**, it tends to lose the information because it cannot store the long sequences and as the methodology is concerned it focuses only on the latest information available at the node.

RNN - Gated Recurrent Unit (GRU)

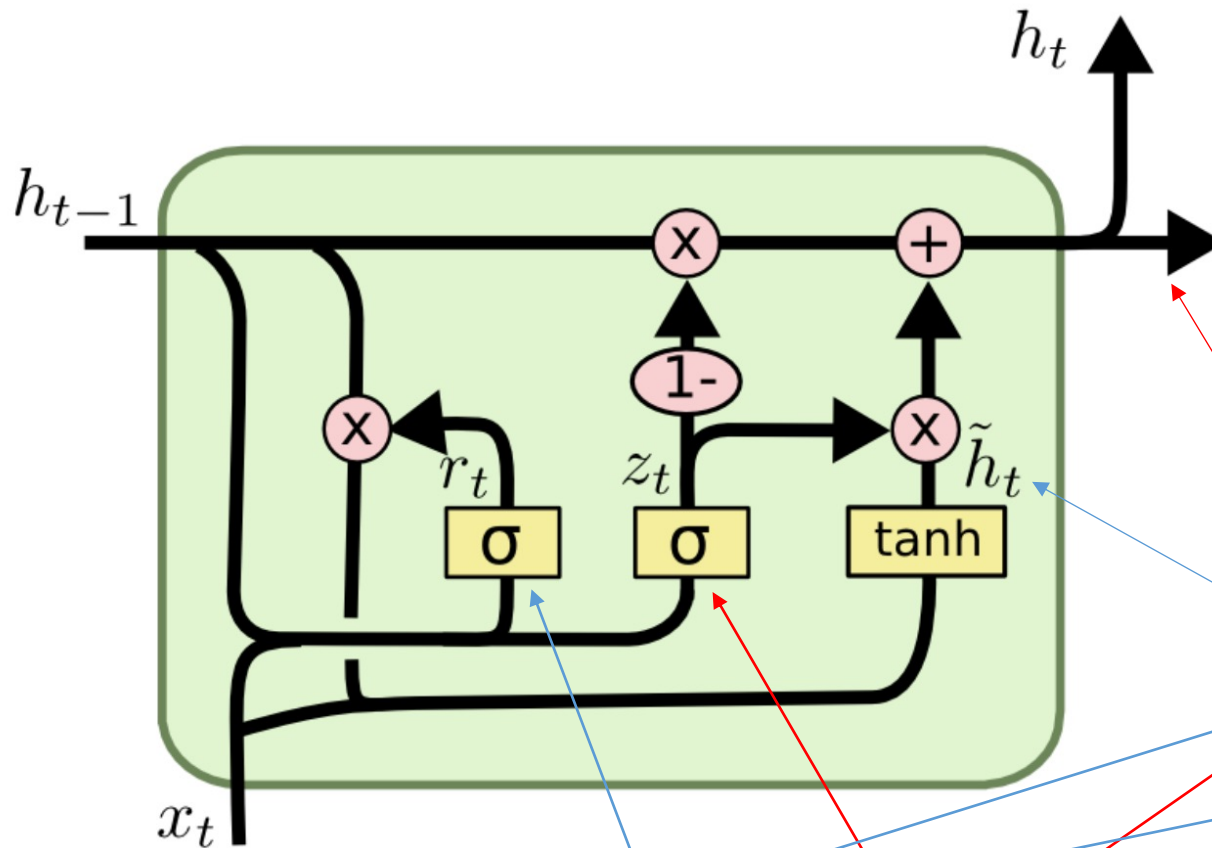
h_t : hidden layer vectors.

x_t : input vector.

b_z, b_r, b_h : bias vector.

W_z, W_r, W_h : parameter matrices.

$\sigma, \tanh$: activation functions.



Gates: $\rightarrow 0 / \rightarrow 1$
Sigmoid function naturally saturates at 0 and 1

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r)$$

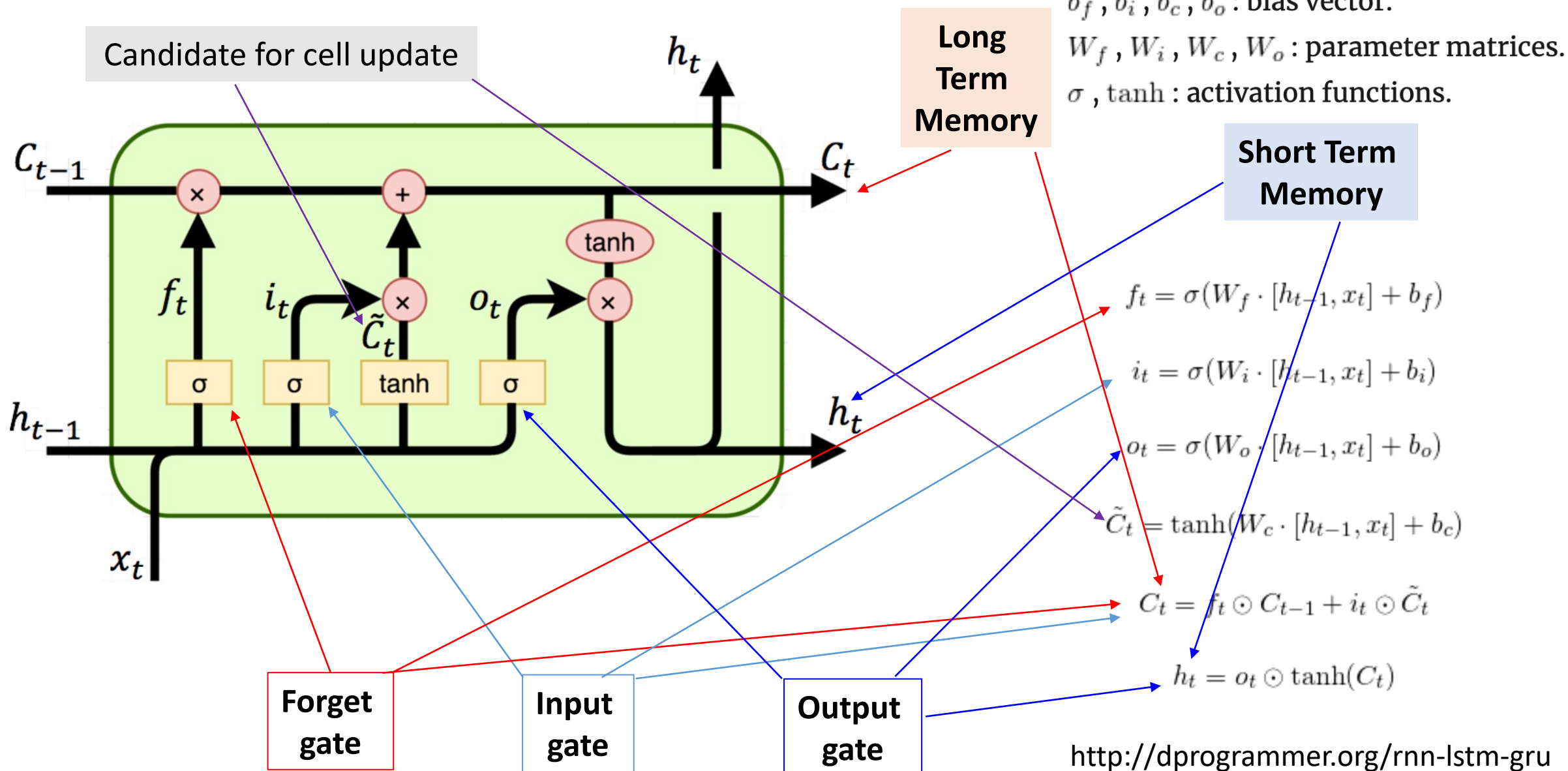
$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

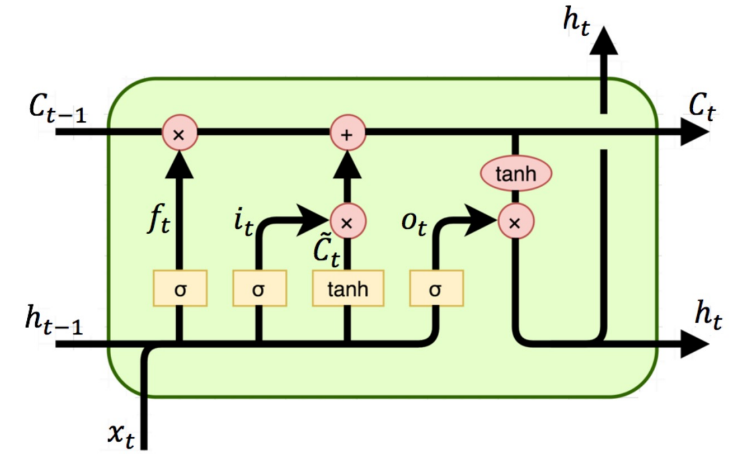
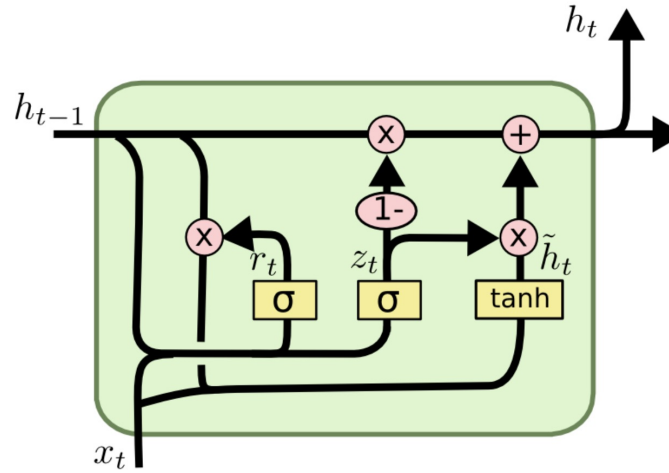
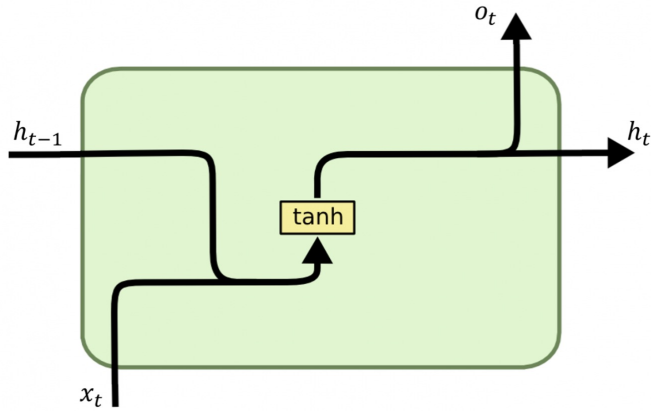
Reset
gate

Update
gate

RNN – Long Short Term Memory (LSTM)



RNN – GRU – LSTM



- **Number of gates:** RNN (no gates) < GRU (2 gates) < LSTM (3 gates)
- **Training difficulty:** RNN < GRU < LSTM (very difficulty to train in fact)
- **Effectiveness:** RNN < GRU $\approx$ LSTM

Small datasets

Large datasets

LSTM Success Stories

LSTMs have been used to win many competitions in speech and handwriting recognition.

Google used LSTM for speech recognition on the smartphone, for Google Translate.

Amazon uses LSTM for Amazon Alexa.

In 2017, Facebook performed some 4.5 billion automatic translations every day using long short-term memory networks

Major technology companies including Google, Apple, and Microsoft are using LSTMs as fundamental components in products.

Apple uses LSTM for the "Quicktype" function on the iPhone and for Siri.

LSTM cannot be trained in parallel...

Transformers!!!

Further Information - Codes

RNN, including GRU and LSTM, implementation:

<https://www.analyticsvidhya.com/blog/2019/01/fundamentals-deep-learning-recurrent-neural-networks-scratch-python/>

<https://towardsdatascience.com/recurrent-neural-networks-by-example-in-python-ffd204f99470>

<https://www.nbshare.io/notebook/249468051/How-To-Code-RNN-and-LSTM-Neural-Networks-in-Python/>

<https://www.tensorflow.org/guide/keras/rnn>

<https://www.freecodecamp.org/news/the-ultimate-guide-to-recurrent-neural-networks-in-python/>

<https://pythonprogramming.net/recurrent-neural-network-deep-learning-python-tensorflow-keras/>

There are indeed quite a lot that are available online written in different programming languages. Feel free to do a search. 😊